

**SIMATS ENGINEERING
CO3-ASSESSMENT TOOL 3**

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1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

Sol:- Interpretation of the Reward Trend

The Reinforcement Learning (RL) agent is trained for multiple episodes, and its reward values gradually increase from **10 to 45**. This indicates that the agent is **learning from its interaction with the environment and improving its decision-making policy over time**.

A reward represents how good or bad an action was in a particular state. In the beginning, the agent has limited knowledge about the environment, so it may select inappropriate actions and receive relatively low rewards. As training continues, the agent learns which actions produce better long-term outcomes. The gradual increase in reward from 10 to 45 indicates that the RL agent is successfully learning from its interactions with the environment. Initially, the agent receives lower rewards because its policy is not well developed.

As training progresses, it learns to select more effective actions, resulting in higher rewards. The 350% improvement in reward demonstrates significant policy improvement. If the reward eventually stabilizes at a high level, it suggests that the policy is approaching convergence. Therefore, the increasing reward trend indicates improved learning performance and increasingly effective decision-making by the agent.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

Solution: In an epsilon-greedy (ϵ -greedy) strategy, the value of ϵ (epsilon) determines how much the agent explores new actions versus exploits the best action it already knows.

- **Exploration:** Trying different actions to discover whether they give better rewards.
- **Exploitation:** Choosing the action that currently has the highest estimated reward.

The ϵ -greedy strategy works approximately as:

With probability $\epsilon \rightarrow$ explore a random action.

- With probability $1 - \epsilon \rightarrow$ exploit the best-known action.

Interpretation of the given results

Epsilon (ϵ)	Exploration	Exploitation	Cumulative Reward
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Epsilon (ϵ)	Exploration	Exploitation	Cumulative Reward
0.1	10%	90%	Highest
0.3	30%	70%	Medium
0.5	50%	50%	Lowest

$$\epsilon = 0.1$$

With $\epsilon = 0.1$, the agent explores only **10%** of the time and exploits the best-known action **90%** of the time.

Therefore, once the agent identifies good actions, it selects them frequently. This results in a **higher cumulative reward**.

$$\epsilon = 0.3$$

Here, the agent explores **30%** of the time and exploits **70%** of the time.

This provides a better balance between exploration and exploitation, but compared with $\epsilon = 0.1$, the agent spends more decisions trying alternative actions. Therefore, the cumulative reward is lower in the given experiment.

$$\epsilon = 0.5$$

With $\epsilon = 0.5$, the agent explores **50%** of the time and exploits only **50%** of the time.

This means the agent frequently chooses random actions, even after learning which actions are good. Consequently, fewer decisions are devoted to the best-known actions, resulting in the **lowest cumulative reward**.

Why does the reward decrease as ϵ increases?

As ϵ increases:

More exploration → More random actions → Less exploitation → Lower cumulative reward

In this particular experiment, the environment apparently allows the agent to identify good actions relatively quickly. Therefore, excessive exploration is unnecessary and reduces the reward.

Best epsilon value

The best value among the tested values is:

$$\epsilon = 0.1$$

because it produced the **highest cumulative reward**.

It provides:

- 10% exploration
- 90% exploitation
- Better use of learned information
- Higher cumulative reward

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm

Solution:Q-Learning and SARSA are both model-free Reinforcement Learning algorithms that learn the value of taking an action in a particular state. However, they differ in how they update their Q-values.

1. Performance difference

The question states that **Q-Learning consistently produces slightly higher average rewards than SARSA** at 100, 200, and 300 episodes.

This means:

- At **100 episodes**, Q-Learning performs slightly better.
- At **200 episodes**, Q-Learning continues to achieve a higher reward.
- At **300 episodes**, Q-Learning still maintains the advantage.
- The difference is described as **slight**, so both algorithms are learning successfully.

The important observation is that the performance gap remains consistent as the number of training episodes increases.

4.An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

Sol: Discount Factor and Long-Term Reward Optimization

The **discount factor (γ)** determines how much importance an RL agent gives to **future rewards** compared with immediate rewards.

The value of γ normally lies between **0 and 1**.

- $\gamma = 0.2 \rightarrow$ Strong preference for immediate rewards.
- $\gamma = 0.5 \rightarrow$ Gives moderate importance to future rewards.
- $\gamma = 0.9 \rightarrow$ Gives high importance to future rewards.

Given observation

The average reward increases as the discount factor increases:

Discount Factor	Importance of Future Rewards	Performance
0.2	Low	Lower reward
0.5	Moderate	Better reward
0.9	High	Highest reward

Interpretation

With $\gamma = 0.2$, the agent mainly focuses on immediate rewards. It may choose an action that provides a quick reward even if that action leads to a poorer long-term outcome.

With $\gamma = 0.5$, the agent considers both immediate and future rewards. Therefore, it can make better long-term decisions.

With $\gamma = 0.9$, future rewards receive much greater importance. The agent is more likely to choose actions that may provide a smaller immediate reward but lead to higher cumulative rewards in the future.

The discounted return can be represented as:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

Therefore, increasing γ causes future rewards to contribute more strongly to the agent's decision-making.

Best discount factor

Based on the given experimental results, $\gamma = 0.9$ is the most suitable value because it produces the highest average reward.

However, $\gamma = 0.9$ is not universally the best value. The appropriate value depends on the environment. For tasks requiring long-term planning, a higher discount factor is generally useful.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

Sol: The success rate increases from **45% at 50 episodes** to **92% at 200 episodes**.

Given data

Training Episodes	Success Rate
50	45%
200	92%

Improvement

The absolute improvement is:

$$92\% - 45\% = 47\%$$

So, the success rate improved by **47 percentage points**.

The relative improvement is:

$$\frac{92 - 45}{45} \times 100 \approx 104.44\%$$

Therefore, the success rate more than **doubled relative to its initial value**.

Interpretation

At **50 episodes**, the success rate is only 45%. This suggests that the agent has not learned the environment sufficiently and is still exploring different actions.

As training continues, the agent gains more experience and learns which actions produce successful outcomes.

At **200 episodes**, the success rate reaches 92%. This indicates that the agent has learned a much more effective policy.

The trend is:

```

50 Episodes
  ↓
45% Success
  ↓
More Training
  ↓
Better Policy
  ↓
200 Episodes
  ↓
92% Success

```

Has the agent learned an optimal policy?

A **92% success rate strongly indicates that the agent has learned an effective policy**, but it does **not by itself prove that the policy is optimal**.

To claim that the policy has converged toward an optimal policy, we should also check:

1. Whether the success rate becomes stable after 200 episodes.
2. Whether additional training produces little or no improvement.
3. Whether the average reward has converged.
4. Whether the agent performs consistently on different initial states.
5. Whether the policy performs well on unseen situations.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

Sol: Learning Rate in CartPole

The **learning rate** determines how much the RL agent changes its model or Q-values after learning from new experience.

Given learning rates:

- **0.01**
- **0.05**
- **0.1**

The agent achieves the **highest stability at 0.05**.

Effect of Learning Rate

Learning Rate	Behavior	Expected Result
0.01	Very small updates	Slow learning and slow convergence
0.05	Moderate updates	Good balance between speed and stability
0.1	Very large updates	Faster changes but possible instability

Learning Rate = 0.01

The agent makes very small updates to its policy or value estimates.

Small updates
↓
Slow learning

↓
Slow convergence

Although learning may be stable, the agent can require many episodes to learn an effective policy.

Learning Rate = 0.05

The agent makes moderate updates.

Moderate updates
↓
Good learning speed
+
Good stability
↓
Better convergence

This provides a good balance between learning quickly and avoiding unstable updates.

Learning Rate = 0.1

The agent makes large updates.

Large updates
↓
Fast changes
↓
Possible overshooting
↓
Unstable learning

The agent may change its policy too aggressively, causing fluctuations in performance.

Best Learning Rate

Based on the given results, **0.05 is the best learning rate.**

It provides the best trade-off between:

- **Learning speed**
- **Convergence**
- **Training stability**

7.A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

Sol: Comparison of Exploration Strategies

The three exploration strategies are:

1. **ϵ -greedy**
2. **Softmax**
3. **Random exploration**

The question states that **ϵ -greedy achieves the highest cumulative reward.**

1. **ϵ -Greedy**

ϵ -greedy balances exploration and exploitation.

With probability ϵ , the agent explores a random action.

With probability $1 - \epsilon$, it chooses the action with the highest estimated value.

For example, if:

$\epsilon = 0.1$

then approximately:

10% $\rightarrow$ Explore

90% $\rightarrow$ Exploit

This allows the agent to discover new actions while mostly selecting actions that it already knows are good.

2. Softmax Exploration

Softmax selects actions according to their estimated values.

Actions with higher Q-values receive higher probabilities, while lower-valued actions receive smaller probabilities.

For example:

Action A $\rightarrow$ 70%

Action B $\rightarrow$ 20%

Action C $\rightarrow$ 10%

Unlike ϵ -greedy, softmax does not simply divide decisions into completely random exploration and greedy exploitation. However, its performance depends heavily on the temperature parameter.

3. Random Exploration

Random exploration selects actions randomly without strongly considering their learned values.

For example:

Action A → Random
Action B → Random
Action C → Random

This provides extensive exploration but does not effectively exploit the knowledge gained by the agent.

Therefore, it generally produces lower cumulative rewards once the agent has learned useful actions.

Comparison

Strategy	Exploration	Exploitation	Expected Performance
ϵ - Greedy	Controlled	Strong	Highest in given scenario
Softmax	Probability-based	Probability-based	Moderate
Random	Very high	Very low	Lowest

Why does ϵ -greedy perform better?

The main reason is its **simple and effective balance between exploration and exploitation**.

During early training, exploration allows the agent to discover useful actions.

As the agent learns, exploitation allows it to repeatedly select high-value actions.

```
Explore
↓
Learn Q-values
↓
Identify good actions
↓
Exploit good actions
↓
Higher cumulative reward
```

Random exploration does not sufficiently exploit learned knowledge, so it wastes many actions on random choices.

Softmax can provide a more gradual action selection strategy, but its performance depends on the temperature parameter. If the temperature is not properly selected, the agent may either explore too much or exploit too aggressively.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

Sol:- Reduction in Steps in Maze Environment

The RL agent initially takes **50 steps** to reach the goal, but after **300 episodes**, it takes only **15 steps**.

Interpretation

The reduction is:

$$50 - 15 = 35 \text{ steps}$$

Percentage reduction:

$$\frac{50 - 15}{50} \times 100 = 70\%$$

Therefore, the number of steps has reduced by **70%**.

This indicates that the agent has learned a more efficient route through the maze.

Initially:

50 steps
↓
Poor/unknown policy
↓
More exploration

After training:

15 steps
↓
Better policy
↓

Efficient navigation

Does this indicate convergence to an optimal policy?

The reduction in steps is a **strong indication of policy improvement**, but it does not alone prove that the policy is optimal.

To confirm convergence, we should check whether:

- The number of steps remains close to 15 for additional episodes.
- The reward becomes stable.
- The agent consistently reaches the goal.
- Further training does not significantly reduce the number of steps.
- The agent performs well from different starting conditions.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

Sol:-SARSA and Discount Factor

The SARSA agent is tested with discount factors:

- **0.4**
- **0.7**
- **0.95**

The highest long-term reward is obtained with **0.95**.

Role of Discount Factor

The discount factor γ (**gamma**) determines how much importance the agent gives to future rewards.

It normally ranges between:

$$0 \leq \gamma < 1$$

The return is calculated as:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

$$\gamma = 0.4$$

The agent gives relatively more importance to immediate rewards.

Immediate reward $\rightarrow$ Important

Future reward → Less important

This can result in short-term decision-making.

$$\gamma = 0.7$$

The agent gives a moderate importance to future rewards.

Immediate rewards
+
Future rewards
↓
Balanced decision-making

$$\gamma = 0.95$$

The agent gives very high importance to future rewards.

Current action
↓
Future consequences
↓
Long-term reward

This is useful when actions taken now influence outcomes many steps later.

Why is 0.95 most effective?

The experiment shows that $\gamma = 0.95$ produces the highest long-term reward.

This indicates that the SARSA agent benefits from considering future consequences when making decisions.

A high discount factor helps the agent avoid selecting an action merely because it gives an immediate reward. Instead, it can choose an action that produces better cumulative rewards over time.

Comparison

Discount Factor	Future Reward Importance	Behavior
0.4	Low	Short-term focus
0.7	Moderate	Balanced
0.95	Very high	Long-term focus

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

Sol:-Fluctuating Rewards and Policy Stability

The RL agent is trained in a **stochastic environment**. Initially, rewards fluctuate, but after **500 episodes**, they become stable.

Interpretation of Early Fluctuations

In the beginning, the agent has limited knowledge about the environment.

It explores different actions:

```
Episode 1 → Reward = 10
Episode 2 → Reward = 5
Episode 3 → Reward = 15
Episode 4 → Reward = 7
Episode 5 → Reward = 18
```

Therefore, the reward can fluctuate significantly.

The stochastic environment also introduces randomness in transitions and outcomes.

After 500 Episodes

As training progresses, the agent learns which actions are generally more beneficial.

The reward may become:

```
Episode 500 → 40
Episode 520 → 42
Episode 540 → 41
Episode 560 → 42
Episode 580 → 41
```

The rewards are now much more stable.

What does stability indicate?

Reward stability indicates that the agent's policy has become more consistent.

```
Early training
  ↓
High exploration
  ↓
Fluctuating rewards
  ↓
Learning
  ↓
```

Better policy
↓
Stable rewards

This suggests:

1. The agent has learned useful behavior.
2. The policy is becoming stable.
3. The agent is less dependent on random actions.
4. The agent is making more consistent decisions.
5. The learning process is becoming robust to environmental randomness.